MYSORE WEST LIONS SEVANIKETAN SCHOOL

Gokulam 3rd Stage, Gokulam, Mysuru, Karnataka 570002

OFFICIAL PROJECT SYNOPSIS

AUTOMATED CLASSROOM NOISE MONITOR

Smart Classroom Decibel Monitor & Principal Alert System

Prepared By:	Saraveswaran N. & Jahnavi S.B.	**Coordinator:**	Rashmi P.S.
Grade / Class:	10th Grade	**Contact No:**	+91 97393 70158
Email Address:	rashandy09@gmail.com		

1. INTRODUCTION & PROBLEM STATEMENT

In educational institutions, the unscheduled absence or brief departure of a classroom teacher often leads to unstructured noise. This unmanaged volume significantly disrupts the academic concentration of neighboring classrooms. Relying on staff for continuous manual monitoring across multiple hallways is highly labor-intensive and administrative-inefficient.

To address this challenge, this project introduces an **automated, IoT-based engineering solution**. The system continuously measures acoustic decibel levels inside the classroom in real time and automatically alerts school administrators or the principal whenever a predetermined noise threshold is breached.

2. WORKING PRINCIPLE & OPERATIONAL MODES

Functioning as an automated digital supervisor, an electronic sound sensor continuously samples ambient decibel levels and converts acoustic waves into digital signals. The system operates across three dynamic operational tiers:

🟢 Normal Mode (Safe Acoustic Baseline)

When noise remains below the preset baseline, the system stays passive, giving a reassuring green light indicator inside the room.

🟡 Warning Mode (Transient Noise Spikes)

When sudden, brief volume spikes occur (e.g., loud talking or laughter), a local warning light flashes to signal students to self-regulate their noise level.

🔴 Alert Mode (Sustained Noise Violation & Escalation)

If classroom volume remains elevated for a continuous duration (e.g., >45 seconds), the system logs a violation and wirelessly transmits an instant alert (SMS/Email/App Notification) directly to the principal's dashboard.

3. SYSTEM COMPONENTS & ARCHITECTURE

CATEGORY	COMPONENT	FUNCTION & CORE ROLE
The 'Ear' (Sensor)	**LM393 / KY-038** Sound Sensor	

CATEGORY	COMPONENT	FUNCTION & CORE ROLE
		Continuously tracks acoustic decibel fluctuations and converts analog sound waves into digital signals.
The 'Brain' (Controller)	**NodeMCU ESP8266** Board	Processes real-time signals, handles delay logic algorithms, and hosts Wi-Fi connectivity.
The 'Voice' (Alert Network)	**IFTTT / Blynk Cloud** Platform	Routes wireless triggers to push notifications, SMS, or email alerts sent directly to school administrators.
Local Feedback	**Red & Green LEDs / Buzzer**	Provides immediate visual and audible feedback to students, prompting self-governance before escalation.

4. KEY FEATURES & EDGE ADVANTAGES

- **Intelligent False-Alarm Filtering:** Built-in code logic filters out transient background sounds (such as a dropped book, moving a chair, or a sneeze). An alert triggers only when elevated noise is sustained over a custom time window.
- **Wireless & Non-Invasive Deployment:** Utilizes existing school Wi-Fi networks, eliminating the need for long, costly, and unsightly wiring across campus corridors.
- **Behavioral Intervention:** Local visual and audio indicators foster student self-discipline, encouraging them to keep noise levels low independently.

5. CONCLUSION & FUTURE SCOPE

This project presents a practical, scalable, and cost-effective application of IoT technology tailored for educational institution management. Future enhancements include integrating a centralized web analytics dashboard to map noise patterns across all classrooms, helping administrators optimize class schedules and identify systemic disciplinary trends.

Student Signature | Project Coordinator | Principal Signature